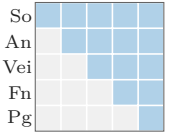
Model

material point (ϕ_i, ψ_i), $i = 1, \ldots, 5$
fraction, viability

couples →



$\mathbf{A} = \mathbf{A}^{\mathsf{T}}$
15 unknowns

energy $\Psi = -\frac{1}{2}\, c^{*}\, \bar{\boldsymbol{\phi}}^{\mathsf{T}} \mathbf{A}\, \bar{\boldsymbol{\phi}}$, $\bar{\phi}_i = \phi_i \psi_i$ (living fraction)

Data

CS commensal static
DS dysbiotic static
CH commensal HOBIC
DH dysbiotic HOBIC

5 species, days 1–21: composition + viability

pH held out for checking

Staged TMCMC

3 seeds per stage; drawn width ∝ prior box

(i) composition initial box → **(ii)** ±4 SD → **(iii)** + viability initial box → **(iv)** ±4 SD

accept a stage only if
(a) $\ln L$ recomputed (b) β reaches 1
(c) ≥5 accepted moves per parameter
(d) $\max \ln L$ within 1 across seeds
(e) medians within $\frac{1}{2}$ SD across seeds
else: more particles or wider box, repeat

each stage: tempering $\beta: 0 \to 1$, MH moves (RW / DE); forward model in JAX, `jit+vmap` on a GPU